Syllabus Details

Syllabus ID:	14181
Syllabus Name:	**SE Graduation Thesis_Khóa luận tốt nghiệp Kỹ thuật phần mềm**
Course Name English:	**SE Graduation Thesis**
Subject Code:	**SET490**
Learning-Teaching Method:	Inquiry-based Teaching, Experiential learning methods, Project Based Training, Research-Based Learning
NoCredit:	10
Degree Level:	Bachelor
Time Allocation:	Study hour (500h) = 33.75h contact hours (1newslot/week*15week) + 1.5h thesis defense + 464.75h self-study
Pre-Requisite:	o Successfully complete (*) below topics: - Project Management – PMG201c or PMG202c - Software Architecture and Design – SWD391 or SWD392 - Software development project - SWP391 o Pass 60% of total credits of selected combo. o Pass OJT202 (not apply to international students) o Pass 80% of total credits (PHE_COM* and OJT excluded) (*): GPA must be greater than or equal to 4.
Description:	SE Graduation Thesis is designed to provide students with essential skills and initial research-oriented experience in participating in real-world software engineering projects. Students are expected not only to apply the knowledge and techniques acquired throughout their program, but also to identify, analyze, and address a research problem within the context of software development. Student can apply knowledges they have learned to the real current projects include: - Experiencing across all project phases: requirement, design, coding, testing and release - Finding out and using the templates and composing documents for each project stage - Using knowledges of technology that have been taught - Having the opportunity to research and apply new technologies - Practicing teamwork and how to interact with colleagues in a project implementation process - Ensuring the proper implementation of quality requirements and schedule - Doing troubleshooting - In addition to applying existing technologies, students are encouraged to explore, evaluate, and integrate new technologies or approaches relevant to their research problem. With the team size of 4-5 members OR individual, the output product size are required to be equivalent to 8-10 medium use cases (3-7 transactions for each use case; action buttons on the screen, end user interactions to the software screens/functions or database transactions can be counted as transaction) At the same time, the project must incorporate a research component, including problem definition, literature review, proposed solution or model, and experimental evaluation. Students are expected to justify their technical decisions based on both practical considerations and existing research. The graduation thesis provides students, working in groups, with a significant project experience in which they can integrate much of the material they have learned in their program, including matters relating to requirements, design, implementation, human factors, professionalism, and project management. Students should be expected to deliver one or several iterations of a software system, along with all artifacts appropriate to the process model they are using. These would likely include a) Research Proposal and Plan(perhaps updated regularly), b) Literature Review/Evaluation c) Methodology Report and Data Collection Results d) Experimental and Results e) Discussion, Conclusion, and Future Work f) Software Package (source code, deployment package, installation guide, user manuals), Paper (including Publication Information- if any) All the documents in the project implementation process and the progress reports are required to write in English.

StudentTasks:	# Choose appropriate project # Form group (if need). # Divide and schedule work (among members). # Attend all scheduled meeting with supervisor. # Produce document and product according to project requirement and member work share. # Defend work infront of examination board, showing both project results and individual responsibility. # In case of cheating found, the whole team is not allow to defend. #Students are required to attend at least 80% of scheduled meetings with their supervisor(s) to be eligible for the thesis or capstone defense.
Tools:	Appropriate tools for design and implementation.
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	377/QĐ-ĐHFPT dated 04/09/2026
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	4/9/2026

3 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Graduated Regulation (Quy che tot nghiep)						True	False	False	Bản mềm, tài liệu xây dựng bởi FU
2	Templates						True	False	False	Bản mềm, tài liệu xây dựng bởi FU
3	The Student's Guide of Graduation Thesis						True	False	False	Bản mềm, tài liệu xây dựng bởi FU

7 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Problem Identification & Research Gap
2	CLO2	Literature Review & Synthesis
3	CLO3	Research Design & Methodology
4	CLO4	Proposed Solution / Model Development
5	CLO5	Implementation & Experimental Evaluation
6	CLO6	Analysis & Critical Discussion
7	CLO7	Research Communication & Integration

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Discussion on the CP topic, CP Structure, Guide Report No.1: Introduction Report No.1 Submission: Project Introduction	Offline	CLO1	ITU	Report 1	SET490	Prepare Report No.1 Guided study: at leat 24 units per week.	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
2	Team Weekly Meeting: CP Topic Review & Group Division Guide Report No.2: Research Proposal and Plan Report No.2 Submission: Project Plan	Offline	CLO1	TU	Report 2	SET490	Prepare Report No.2 Guided study: at leat 24 units per week.	
3	Team Weekly Meeting: Project Plan Review Guide Report No.3: Literature Review/Evaluation Report No.3 Submission: Literature Review/Evaluation	Offline	CLO2, CLO3	TU	Report 3		Prepare Report No.3 Update Project Introduction, Project Plan Guided study: at leat 24 units per week.	
4	Team Weekly Meeting: Project & Methodology Report and Data Collection Results Team Weekly Meeting: Methodology Report and Data Collection Results Report No.4 Submission: Methodology Report and Data Collection Results	Offline	CLO2, CLO3, CLO4	TU	Report 4		Prepare Report No.4 Update Project Introduction, Project Plan, and Methodology Report and Data Collection Results Guided study: at leat 24 units per week.	
5	cont.	Offline	CLO2, CLO3, CLO4	TU				
6	Team Weekly Meeting: Literature Review/Evaluation Team Weekly Meeting: Code interation1 Product Demo	Offline	CLO2, CLO3, CLO4	TU	Report 4		Prepare Report No.4 Update Project Introduction, Project Plan, and Methodology Report and Data Collection Results Guided study: at leat 24 units per week.	
7	cont.	Offline	CLO2, CLO3, CLO4	TU				
8	Team Weekly Meeting: Methodology Report and Data Collection Results Team Weekly Meeting: Code Iteration 2 Product Demo	Offline	CLO2, CLO3, CLO4	TU	Report 4		Prepare Report No.4 Update Project Introduction, Project Plan, and Methodology Report and Data Collection Results Guided study: at leat 24 units per week.	
9	cont.	Offline	CLO2, CLO3, CLO4	TU				
10	Team Weekly Meeting: Code Iteration 3 Plan & Guide Team Weekly Meeting: Code Iteration 3 Product Demo	Offline	CLO2, CLO3, CLO4, CLO5	TU	Report 5		Prepare Report No.5 Update Project Plan, Research Proposal, Literature Review/Evaluation, Methodology Report and Data Collection Results, Experimental and Results Guided study: at leat 24 units per week.	
11	cont.	Offline	CLO2, CLO3, CLO4, CLO5	TU				

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
12	Team Weekly Meeting:Experimental and Results Team Weekly Meeting: Final Package Product Demo Report No.5 Submission: Experimental and Results	Offline	CLO2, CLO3, CLO4, CLO5	TU	Report 5		Prepare Report No.5 Update Project Plan, Research Proposal, Literature Review/Evaluation, Methodology Report and Data Collection Results, Experimental and Results Guided study: at leat 24 units per week.	
13	cont.	Offline	CLO2, CLO3, CLO4, CLO5	TU				
14	Team Weekly Meeting: Project & Software Package Review Guide Report No.6:Discussion, Conclusion, and Future Work Report No.6 Submission: Discussion, Conclusion, and Future Work	Offline	CLO2, CLO3, CLO4, CLO5, CLO6	TU	Report 6		Prepare Report No.6 Guided study: at leat 24 units per week.	
15	Team Weekly Meeting: Practice for Presentation of Capstone Project Report No.7 Submission: Final Project Report, includes all the final project documents (completed Reports No. 1-6, both soft copy & hard copy), Software Product, and thesis presentation slide	Offline	CLO2, CLO3, CLO4, CLO5, CLO6, CLO7	TU	Report 7		Prepare Final Report & Software Package Preparation for Final Presentation Guided study: at leat 24 units per week.	

0 Constructive question(s)

2 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	On-going Assessment	on-going	7	50.0%	5	During the CP Progress	All LOs	Report	Topic of CP		On-going Assessment (OGA): - Report1_Project Introduction: 5% - Report2_Research Proposal and Plan: 5% - Report3_Literature Review/Evaluation: 15% - Report4_Methodology Report and Data Collection Results: 25% - Report 5_Experimental and Results: 25% - Report6_Discussion, Conclusion, and Future Work: 15% - Report7_Final Project Report: 10%	require grade >=2 for each part

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
2	Final Thesis Presentation:	Final exam	1	50.0%	4	~ 30 minutes for presentation, ~ 60 minutes for demo and Q&A. Total is 90 minutes.	All LOs		CP Thesis Presentation		Thesis Defense Assessment (TDA): -Project Introduction: 5% - Research Proposal and Plan: 5% - Literature Review/Evaluation: 15% - Methodology Report and Data Collection Results: 25% - Experimental and Results: 25% - Discussion, Conclusion, and Future Work: 10% - Presentation Skills: 5% - Q&A: 10% Presentation mark is given by each member of defence committee for each student on final project presentation. Personal mark is the average of all marks given by members of defence committee.	One or more specific members of the student team is not allowed to join the thesis defense (and therefore being failed the subject) if he/she has OGA less than 5 or any cheating found from him/her (he/she can't prove they do the software product by him/herself). Before thesis defense, the student team must wait until the second round of thesis defense if the whole team has the OGA less than 5/10 OR they are in one of below cases: Case 1 – Incomplete Research Components The team fails to complete essential research components, including problem definition, literature review, methodology, or experimental evaluation. Case 2 – Insufficient Literature Review The literature review lacks sufficient academic sources (e.g., fewer than the required number of peer-reviewed

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												papers) or fai to provide critical analys and comparison. Case 3 – Lack of Research Contribution The project does not demonstrate a clear research contribution (e.g., no proposed model, algorithm, framework, or meaningful improvement over existing solutions). Case 4 – Missing or Invalid Experimental Results The team fails to conduct experiments, or the results are invalid du to improper methodology, lack of datase or absence of evaluation metrics. Case 5 – Poor Implementatio Quality The develope system is incomplete, non-functiona or does not reflect the proposed research solution. Case 6 – Wea Analysis and Discussion The team fails to analyze results critically, lack justification o findings, or does not discuss limitations and implications.

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												Case 7 – Failure to Submit Required Documents Case 8 – Lack of Traceability Between Research and System There is no clear linkage between the proposed research contribution and the implemented system. Case 9 – Academic Integrity Violation There is considered to occur when similarity exceeds 30% including any instances of plagiarism, falsification o data, or the use of externa sources without appropriate citation. Case 10 – Inadequate Presentation Readiness The team is unable to clearly preser and defend their work, including explaining the research problem, methodology, and results. The student(s has to re-do the capstone project if the above cases occur with the assessments for the secon thesis defens OR there is or

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
												OGA with grade less tha 2/10. After thesis defense, one or more team members are evaluated as failed the capstone project if they are in one of below cases (no matter when the guiding teacher doesn't evaluate that) (1) Student can't demonstrate any function o use case (2) Students can't prove th fact that they prepare any o the project reports or the output software by themselves (3) There are more than 3 obvious coding-logic defects OR more than 1 show-stopper defects (the defects that prevent end user from completing his/her use case on any software functions) in the submitted software.